SAFETY DATA SHEET

1. PRODUCT AND COMPANY IDENTIFICATION

1.1 Product identifiers

Product name : **Methylene Blue Solution, 0.01 M,**
Product number(s):

1.2 Details of the supplier of the safety data sheet

Transene Company, Inc.
1127 57th Avenue
Oakland, CA 94621
USA

Telephone: +1 510-534-1186

1.3 Emergency telephone number

Chemtrec: +1 800-424-9300

2. HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

GHS Classification in accordance with 29 CFR 1910 (OSHA HCS)
Not a hazardous material

2.2 GHS Label elements, including precautionary statements

Pictogram(s)

Signal word **Warning**

Hazard statement(s)
H315- May cause skin irritation
H317- May cause allergic skin reaction
H318- May cause eye irritation

Precautionary statements(s)
P305+P351+P358 -- If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Wear eye protection/face protection

3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1

Name	CAS number	%	GHS–US classification
Methylene Blue	61-73-4	< 1%	Acute Tox. 4 (Oral), H302; Aquatic Acute 2, H401
Water	7732-18-5	balance	N/A

4. FIRST AID MEASURES

4.1 Description of first aid measures

General advice
Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).

If inhaled
Assure fresh air breathing. Allow the victim to rest.

In case of skin contact
Remove affected clothing and wash all exposed skin area with mild soap and water, followed by warm water rinse.

In case of eye contact
Rinse immediately with plenty of water. Obtain medical attention if pain, blinking or redness persist.

If swallowed
Rinse mouth. Do NOT induce vomiting. Obtain emergency medical attention.

4.2 Most important symptoms and effects, both acute and delayed
Not expected to present a significant hazard under anticipated conditions of normal use.

4.3 Indication of any immediate medical attention and special treatment needed None

5. FIREFIGHTING MEASURES

5.1 Suitable extinguishing media
Foam. Dry powder. Carbon dioxide. Water spray. Sand. Do not use a heavy water stream.

5.2 Special hazards arising from the substance or mixture

Hazards
Fire hazard : Not flammable.
Explosion hazard : Not applicable.
Reactivity : None.

5.3 Advice for firefighters

Advice
Use water spray or fog for cooling exposed containers. Exercise caution when fighting any chemical fire. Prevent fire-fighting water from entering environment. Do not enter fire area without proper protective equipment, including respiratory protection.

6. ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures
For non-emergency personnel
Emergency procedures : Evacuate unnecessary personnel.

For emergency responders
Protective equipment : Equip cleanup crew with proper protection.
Emergency procedures : Ventilate area.

6.2 Environmental precautions
Prevent entry to sewers and public waters. Notify authorities if liquid enters sewers or public waters.

6.3 Methods and materials for containment and cleaning up
Soak up spills with inert solids, such as clay or diatomaceous earth as soon as possible. Collect spillage. Store away from other materials.

6.4 Reference to other sections
See Heading 8. Exposure controls and personal protection.

7. HANDLING AND STORAGE

7.1 Precautions for safe handling
Wash hands and other exposed areas with mild soap and water before eating, drinking or smoking and when leaving work. Provide good ventilation in process area to prevent formation of vapour.

7.2 Conditions for safe storage, including any incompatibilities

Storage conditions : Keep container closed when not in use.
Incompatible products : Strong bases. Strong oxidizers. Strong reducing agents.
Incompatible materials : Sources of ignition. Direct sunlight.

7.3 Specific end user(s)
No additional information available

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameter

Components with workplace control parameters

Component	CAS-No.	Control parameters	Value

8.2 Exposure controls

Appropriate Engineering controls
Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure.

Personal protective equipment

Eye/face protection
Chemical goggles or safety glasses.

Skin and Hand protection
Wear protective gloves.

Body protection
Avoid all unnecessary exposure.

Respiratory protection
Wear appropriate mask.

Other information
Do not eat, drink or smoke during use.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Physical state:	Amber Liquid
Color	clear blue
Upper/lower flammability or explosive limits	No data available
Odor	None
Odor threshold	No data available
Vapor pressure	Essentially same as water
Vapor density	Essentially same as water
pH	acidic
Relative density	ca. 1.0
Melting point	ca. 0 C
Freezing point	No data available
Initial Boiling point and boiling range	ca. 100 C
Flash point	No data available
Evaporation rate	Essentially same as water
Flammability (solid, gas)	No data available
Water solubility	soluble
Partition coefficient: n-octanol/water	No data available
Auto-ignition temperature	No data available
Decomposition temperature	No data available
Viscosity	No data available

10. STABILITY AND REACTIVITY

10.1 Reactivity
none

10.2 Chemical stability
Stable under normal conditions.

10.3 Possibility of hazardous reactions
Not established.

10.4 Conditions to avoid
Direct sunlight. Extremely high or low temperatures.

10.5 Incompatible materials
Strong oxidizers. Strong reducing agents. Strong bases.

10.6 Hazardous decomposition products
Nitrogen oxides. Carbon monoxide. Carbon dioxide. Sulfur compounds.

11. TOXICOLOGICAL INFORMATION 11.1

Information on toxicological effects

Acute toxicity
Not classified

Methylene Blue (61-73-4)
LD50 oral rat 1180 mg/kg (Rat)
ATE US (oral) 1180 mg/kg bodyweight

Water (7732-18-5)
LD50 oral rat = 90000 mg/kg
ATE US (oral) 90000 mg/kg bodyweight

Skin corrosion/irritation
Not classified

Serious eye damage/eye irritation
Not classified

Respiratory or skin sensitization
Not classified

Germ cell mutagenicity
Not classified

Carcinogenicity
Not classified

Reproductive toxicity
Not classified

Specific target organ toxicity – single exposure
Not classified

Specific target organ toxicity – repeated exposure
Not classified

Aspiration hazard
Not classified

12. ECOLOGICAL INFORMATION

12.1 Toxicity
Methylene Blue (61-73-4)
LC50 fishes 1 13 mg/l (48 h; Oryzias latipes)

EC50 Daphnia 1 2.26 mg/l (48 h; Daphnia magna)
LC50 fish 2 18 mg/l (96 h; Mystus vittatus)
EC50 Daphnia 2 4.93 mg/l (24 h; Daphnia magna)
TLM fish 1 10 - 100,48 h; Poecilia reticulata

12.2 Persistence and degradability

Methylene Blue, 0.1% Aqueous
Persistence and degradability Not established.

Methylene Blue (61-73-4)
Persistence and degradability Biodegradability in water: no data available. Photodegradation in the air.

Water (7732-18-5)
Persistence and degradability Not established.

12.3 Bioaccumulative potential

Methylene Blue, 0.1% Aqueous
Bioaccumulative potential Not established.

Methylene Blue (61-73-4)
Log Pow 5.85 (Estimated value)
Bioaccumulative potential Not bioaccumulative.

Water (7732-18-5)
Bioaccumulative potential Not established.

12.4 Mobility in soil

No additional information available

12.5 Results in PBT and vPvB assessment

No data available

12.6 Other adverse effects

Avoid release to the environment.

13. DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Product
Dispose in a safe manner in accordance with local/national regulations.

Contaminated Container
Avoid release to the environment.

14. TRANSPORT INFORMATION

In accordance with DOT

DOT Proper Shipping Name Not regulated by D.O.T.

Hazard Labels:

15. REGULATORY INFORMATION

TSCA Inventory List:
All of the ingredients (or their hydrate forms) are listed.
CERCLA Reportable Quantity (RQ):

None of the ingredients have an RQ.
SARA Section 302 Threshold Planning Quality (TPQ):
None of the ingredients have a TPQ.
SARA Title Codes:
CAS # 61-73-4: acute, chronic.
OSHA:
This product is not considered to be highly hazardous by OSHA Hazard Communication Standard.

16. OTHER INFORMATION

HMIS Rating
HEALTH: 1
FIRE: 0
REACTIVITY: 0
SPECIFIC HAZARD: N/A

Further Information

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Preparation Information

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